Subtracting Across Zeros

No tens to break? Think of the number as tens and ones.

403 – 157: 3 – 7? Break a ten. But there are 0 tens!



4 hundreds, 3 ones

Break a hundred into 10 tens, then a ten into 10 ones.

The shortcut: 403 is 40 tens and 3 ones

403 = 40 tens + 3 ones

403 = 39 tens + 13 ones

Take 1 ten from the 40 tens. Now there are 39 tens and 13 ones. One break!

Now subtract: 157 is 15 tens and 7 ones

tens 39 – 15 = 24 tens

ones 13 – 7 = 6

total 240 + 6 = 246

Check by adding:

246 + 157 = 403

Your turn!

302 – 145 = 157

break 302 = 29 tens + 12 ones

tens 29 – 14 = 15

ones 12 – 5 = 7

total 150 + 7 = 157

For teachers

Goal: subtract when the top number has zeros (2.NBT.7). Say "403 is 40 tens and 3 ones," then break one ten: 39 tens and 13 ones. Watch for 0 – 7 = 7 (the 0 can't give). Check every answer by adding.

Subtracting Across Zeros: Practice

Think of the top number as tens and ones. Break once.

Part 1 Tens and ones Write it as tens and ones, then break one ten.

1) 403 = 40 tens + 3 ones
403 = 39 tens + 13 ones

2) 702 = 70 tens + 2 ones
702 = 69 tens + 12 ones

3) 305 = 30 tens + 5 ones
305 = 29 tens + 15 ones

4) 600 = 60 tens + 0 ones
600 = 59 tens + 10 ones

5) 801 = 80 tens + 1 ones
801 = 79 tens + 11 ones

6) 506 = 50 tens + 6 ones
506 = 49 tens + 16 ones

7) 900 = 90 tens + 0 ones
900 = 89 tens + 10 ones

8) 204 = 20 tens + 4 ones
204 = 19 tens + 14 ones

Part 2 Break, then subtract The break row is done for you.

9) 402 − 135 = 267

break	402 = 39 tens + 12 ones
tens	39 − 13 = 26
ones	12 − 5 = 7
total	260 + 7 = 267

10) 703 − 248 = 455

break	703 = 69 tens + 13 ones
tens	69 − 24 = 45
ones	13 − 8 = 5
total	450 + 5 = 455

Subtracting Across Zeros: Practice (continued)

Part 2 Break, then subtract (continued)

11) **500 − 164 = 336**

break 500 = 49 tens + 10 ones

tens 49 − 16 = 33

ones 10 − 4 = 6

total 330 + 6 = 336

12) **610 − 253 = 357**

break 610 = 60 tens + 10 ones

tens 60 − 25 = 35

ones 10 − 3 = 7

total 350 + 7 = 357

Part 3 Fill in every box

13) **804 − 376 = 428**

break 804 = 79 tens + 14 ones

tens 79 − 37 = 42

ones 14 − 6 = 8

total 420 + 8 = 428

14) **301 − 127 = 174**

break 301 = 29 tens + 11 ones

tens 29 − 12 = 17

ones 11 − 7 = 4

total 170 + 4 = 174

15) **900 − 465 = 435**

break 900 = 89 tens + 10 ones

tens 89 − 46 = 43

ones 10 − 5 = 5

total 430 + 5 = 435

16) **520 − 148 = 372**

break 520 = 51 tens + 10 ones

tens 51 − 14 = 37

ones 10 − 8 = 2

total 370 + 2 = 372

Subtracting Across Zeros: Practice (continued)

Part 4 Stack it Cross out the tens, write the new tens and ones on top, then subtract.

17)

hundreds	tens	ones
4	9	13
~~5~~	~~0~~	~~3~~
− 2	7	6
2	2	7

18)

hundreds	tens	ones
3	9	10
~~4~~	~~0~~	~~0~~
− 2	1	8
1	8	2

19)

hundreds	tens	ones
6	12	10
~~7~~	~~3~~	~~0~~
− 3	6	4
3	6	6

20)

hundreds	tens	ones
8	9	11
~~9~~	~~0~~	~~1~~
− 4	5	2
4	4	9

Part 5 Zero in the tens No help this time.

21)

3	9	13
~~4~~	~~0~~	~~3~~
− 1	5	7
2	4	6

22)

5	9	15
~~6~~	~~0~~	~~5~~
− 2	3	8
3	6	7

23)

7	9	12
~~8~~	~~0~~	~~2~~
− 4	6	5
3	3	7

24)

6	9	14
~~7~~	~~0~~	~~4~~
− 3	2	9
3	7	5

25)

4	9	11
~~5~~	~~0~~	~~1~~
− 2	8	6
2	1	5

26)

8	9	16
~~9~~	~~0~~	~~6~~
− 5	4	7
3	5	9

27)

2	9	18
~~3~~	~~0~~	~~8~~
− 1	5	9
1	4	9

28)

1	9	17
~~2~~	~~0~~	~~7~~
− 1	4	8
	5	9

Part 6 Zero in the ones No help this time.

29)

4	10	10
~~5~~	~~1~~	~~0~~
− 2	3	4
2	7	6

30)

6	11	10
~~7~~	~~2~~	~~0~~
− 3	8	5
3	3	5

31)

5	13	10
~~6~~	~~4~~	~~0~~
− 1	7	2
4	6	8

32)

7	12	10
~~8~~	~~3~~	~~0~~
− 4	5	6
3	7	4

Subtracting Across Zeros: Practice (continued)

Part 6 Zero in the ones (continued)

33)

	3	14	10
	~~4~~	~~5~~	~~0~~
−	1	8	7
	2	6	3

34)

	8	10	10
	~~9~~	~~1~~	~~0~~
−	5	3	8
	3	7	2

35)

	2	16	10
	~~3~~	~~7~~	~~0~~
−	1	9	4
	1	7	6

36)

	1	15	10
	~~2~~	~~6~~	~~0~~
−	1	7	5
		8	5

Part 7 Two zeros No help this time.

37)

	2	9	10
	~~3~~	~~0~~	~~0~~
−	1	2	8
	1	7	2

38)

	4	9	10
	~~5~~	~~0~~	~~0~~
−	2	4	6
	2	5	4

39)

	6	9	10
	~~7~~	~~0~~	~~0~~
−	3	5	1
	3	4	9

40)

	3	9	10
	~~4~~	~~0~~	~~0~~
−	1	6	3
	2	3	7

41)

	7	9	10
	~~8~~	~~0~~	~~0~~
−	4	7	2
	3	2	8

42)

	5	9	10
	~~6~~	~~0~~	~~0~~
−	2	8	9
	3	1	1

43)

	8	9	10
	~~9~~	~~0~~	~~0~~
−	5	3	7
	3	6	3

44)

	1	9	10
	~~2~~	~~0~~	~~0~~
−	1	4	6
		5	4

Part 8 Mixed Break only when you need to.

45)

	6	0	5
−	3	0	2
	3	0	3

46)

	7	9	13
	~~8~~	~~0~~	~~3~~
−	2	6	7
	5	3	6

47)

		3	10
	5	~~4~~	~~0~~
−	3	1	5
	2	2	5

48)

	6	9	10
	~~7~~	~~0~~	~~0~~
−	4	2	5
	2	7	5

Subtracting Across Zeros: Practice (continued)

Part 8 Mixed (continued)

49)
$$\begin{array}{r} {\scriptstyle 3\ \ 9\ \ 16} \\ \cancel{4}\ \cancel{0}\ \cancel{6} \\ -\ 1\ 3\ 8 \\ \hline 2\ 6\ 8 \end{array}$$

50)
$$\begin{array}{r} {\scriptstyle 8\ \ 16} \\ \cancel{9}\ \cancel{6}\ 0 \\ -\ 5\ 8\ 0 \\ \hline 3\ 8\ 0 \end{array}$$

51)
$$\begin{array}{r} {\scriptstyle 2\ \ 9\ \ 12} \\ \cancel{3}\ \cancel{0}\ \cancel{2} \\ -\ 1\ 8\ 5 \\ \hline 1\ 1\ 7 \end{array}$$

52)
$$\begin{array}{r} {\scriptstyle \ \ 4\ \ 10} \\ 8\ \cancel{5}\ \cancel{0} \\ -\ 6\ 2\ 4 \\ \hline 2\ 2\ 6 \end{array}$$

Part 9 Check by adding No help this time. Do you get the top number back?

53) 246 + 157 = 403

54) 177 + 128 = 305

55) 365 + 245 = 610

56) 226 + 274 = 500

57) 318 + 282 = 600

58) 459 + 141 = 600

Part 10 Remember? Problems from earlier packets.

59) 65¢ + 25¢ = 90¢

60) half past 7 = 7:30

61) 352 + 216 = 568

62) 64 + 36 = 100

63) half of 86 = 43

64) 2 halves = 1 whole

Part 11 Word problems Break once. Check by adding.

65) A school has 400 pencils. The kids use 156 of them. How many pencils are left?

Number sentence 400 − 156 = 244

Answer 244 pencils

Draw or show your work

Subtracting Across Zeros: Practice (continued)

Part 11 Word problems (continued)

66) A book has 305 pages. Mia has read 128 pages. How many pages are left to read?

Number sentence 305 − 128 = 177

Answer 177 pages

Draw or show your work

67) There were 610 people at a game. Then 245 people went home. How many people stayed?

Number sentence 610 − 245 = 365

Answer 365 people

Draw or show your work

68) A jar holds 500 beads. 274 beads are red and the rest are blue. How many beads are blue?

Number sentence 500 − 274 = 226

Answer 226 beads

Draw or show your work

69) Challenge: A game needs 1000 points. Kim has 455. How many more points does she need?

Number sentence 455 + 545 = 1000

Answer 545 points

Draw or show your work

Check-In: Subtracting Across Zeros

Show what you know. No help this time.

Mastery: 8 of 9 (80%)

Part 1 Tens and ones

1) 903 = 90 tens + 3 ones

903 = 89 tens + 13 ones

2) 409 = 40 tens + 9 ones

409 = 39 tens + 19 ones

Part 2 Subtract Break once.

3)

	5	10	10
	~~6~~	~~1~~	~~0~~
−	4	7	3
	1	3	7

4)

	7	9	12
	~~8~~	~~0~~	~~2~~
−	2	1	3
	5	8	9

5)

	8	16	10
	~~9~~	~~7~~	~~0~~
−	6	8	1
	2	8	9

6)

	7	12	10
	~~8~~	~~3~~	~~0~~
−	7	3	3
		9	7

Part 3 Check by adding

7) 268 + 632 = 900

8) 85 + 615 = 700

9) 464 + 356 = 820